

Data Mining and Machine Learning

Intermediate Results Report

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Project Title:

Early Prediction of Customer Escalation in Support Tickets Using Context-Aware Retrieval-Augmented Analytics

1. Project Overview

The project investigates the issue of whether or not support tickets that will escalate can be identified prior to them escalating. Escalation is not a sudden occurrence in most real support environments. Rather, it accumulates with time, due to recurring confrontations, unfinished business or the mounting customer dissatisfaction.

Majority of the current available methods are inclined towards analysing the text of the tickets or identifying negative sentiment. These techniques may be effective but they tend to identify signals when the situation has already gone bad. That restricts their application in practice where a more timely intervention will be more useful.

This project has a bit of a different approach. Instead of considering each ticket separately, the concept is to utilize patterns of previous tickets. Similar issues in support systems have a tendency to take similar paths. When it is possible to identify those patterns early, they can possibly be used to predict escalation risk.

The primary tasks at this place of the project have been to establish a solid data pipeline, learn about the data with exploratory analysis, and to prepare baseline machine learning models. These findings will serve as a benchmark to analyze the more sophisticated methods at the second stage, especially retrieval-based context and sentiment progression.

2. Dataset Description

In order to create a realistic support environment, several datasets were merged as opposed to using a single source. The data sets are:

- Customer care email conversations (https://huggingface.co/datasets/rtweera/customer_care_emails)
- Synthetic IT support tickets (<https://huggingface.co/datasets/KameronB/synthetic-it-callcenter-tickets>)
- CRM-based support ticket data (<https://www.kaggle.com/datasets/ajverse/customer-support-tickets-crm-dataset>)

After combining the available datasets, the working dataset consisted of approximately **29,861 records**.

A primary issue at this point was that the structure of each dataset was different. Others had more in-depth conversation threads whereas others had other useful metadata like satisfaction scores or resolution notes. Due to this, a common schema was required to be developed where all the datasets could be utilized.

Each ticket was standardised into a format that includes:

- A unique ticket identifier
- The full conversation text
- A derived sentiment score
- A binary escalation label

This step was important to ensure consistency across the pipeline.

3. Data Processing and Preparation

3.1 Data Loading

The data loading process was designed in a modular way to allow separate handling of multiple datasets before combining them into a unified structure. In this project, two datasets were sourced from Hugging Face and one from Kaggle. Since these datasets had slightly different formats and structures, some preprocessing steps were required to align them before merging. Despite these differences, the datasets were successfully integrated, providing sufficient data for further experimentation and analysis.

3.2 Preprocessing

The preprocessing phase was aimed at ensuring that the data is made usable without losses of significant information. Cleaning of the text was maintained to a minimum. Rather than such violent changes as stemming or heavy filtering, it was simply aimed at getting rid of noise, such as unnecessary whitespace and irrelevant characters, without altering the original meaning. The missing values were treated with caution to prevent biasness. Subsequently, SMOTE was used to deal with class imbalance. This was needed since the data was skewed towards a certain type which might influence the learning of models.

After preprocessing:

- Total samples: **5,504**
- Balanced distribution: **2,752 escalated and 2,752 non-escalated**

Balancing the dataset made a noticeable difference in model behaviour during training.

3.3 Exploratory Data Analysis (EDA)

The data was pre-modelling with EDA in order to get a better understanding of the data behaviour. A number of visualisations were developed, such as:

- Escalated vs non-escalated tickets distribution.
- Distribution of text length
- Sentiment score distribution
- Word frequency plots and word clouds.

Another curious finding was that escalated tickets have a higher likelihood of a more protracted discussion. This intuitively makes sense, since unsolved problems tend to need additional back and

forth communication. The other result was that sentiment has no clear distinction between escalated and non-escalated tickets. The overlap is substantial, which indicates that sentiment may not be adequately addressed by looking at it separately. This encourages the notion of examining how sentiment varies with time as opposed to a single score. Even some problems in this step were encountered especially when plotting functions. There were also some mistakes, which were brought about by wrong data formats but were rectified after debugging and modifying the implementation.

4. Baseline Model Implementation

Three machine learning models were trained on TF-IDF features on the text in the ticket to set a starting point:

- Logistic Regression
- Random Forest
- XGBoost

The dataset was split into training and test sets using a standard approach, ensuring that both sets maintained the same class distribution.

These models were chosen to represent different types of learning:

- Linear model (Logistic Regression)
- Ensemble model (Random Forest)
- Boosting model (XGBoost)

This allows for a more meaningful comparison of performance.

5. Model Results

Logistic Regression

- F1 Score: 0.5984
- ROC-AUC: 0.6368

The performance of this model was rather modest. Convergence warnings also witnesses during training and this is an indication that the model was not able to obtain an optimal solution within specified iterations.

Random Forest

- F1 Score: 0.7215
- ROC-AUC: 0.7276

Random Forest performed significantly better than Logistic Regression. It handled the complexity of the data more effectively, especially in capturing non-linear relationships.

XGBoost

- F1 Score: 0.7135
- ROC-AUC: 0.7319

XGBoost achieved the best ROC-AUC score among the three models, indicating stronger overall classification capability. Its performance was slightly more stable compared to Random Forest.

6. Discussion of Results

The findings bring to the fore some noteworthy issues.

First, this problem is not well addressed by the simpler linear models. The correlation among features seems to be more difficult, and this is why tree-based approaches are more effective.

Second, although the models score reasonably, there are still a significant amount of false predictions, and false negatives. This is an issue since the inability to notice a possible escalation will decrease the utility of the system.

Third, TF-IDF features alone restrict the amount of context that the models can learn. The models are basically basing their attention on the superficial text patterns, without knowing how similar tickets performed earlier or how the discussion is developing.

These limitations are leading to the subsequent stage of the project.

7. Next Steps

The next stage will focus on improving the model by introducing additional sources of information.

The main areas of work are:

- Building a retrieval system to identify similar historical tickets
- Extracting structured features from those tickets
- Modelling how sentiment changes across a conversation instead of treating it as static
- Retraining and comparing models using these enhanced features

The goal is to determine whether these additions lead to more reliable and earlier prediction of escalation.